

Lecture #1: parallelism basics within one GPU

outlines/goals:

- #1: understand the systems complexities of training huge models
- #2: Different parallelization paradigms and why people use
- #3: what large scale training runs multiple at once often look like;

part 1: Basics of networking for LLMs.

part 2: Different forms of parallel LLM training

part 3: scaling and training big LLMs with parallelism

↓ limits to ^{single} GPU-based scaling - compute.

A single GPU can't fit most LLMs, as they get larger

↓ High intra-node (within one machine) parallelism via high-speed interconnects

↓ NVSwitch = GPU connections

High inter-node ^{→ NVlink} (across multiple machine) parallelism.

HDR Intra Band

50 GT/s

PCI expr 4.0

16 GT/s

XGMI-2

16 GT/s

NVlink 3.0

400 GT/s per lane

collective communication: $\rightarrow$ one rank is one GPU (process)

- all reduce

Reduce

Broadcast $\rightarrow$ one GPU sends

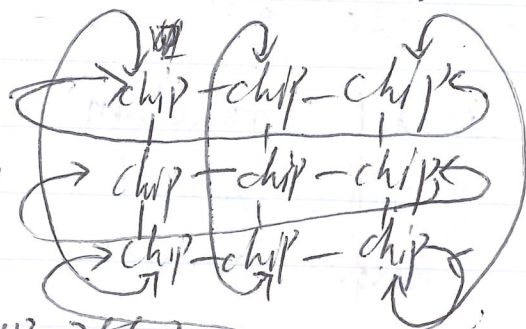
All Gather

Reduce Scatter

$$\Rightarrow \text{All Reduce} = [\text{Reduce} + \text{scatter}] + \text{All gather}$$

$\rightarrow$ Connect with all TPU

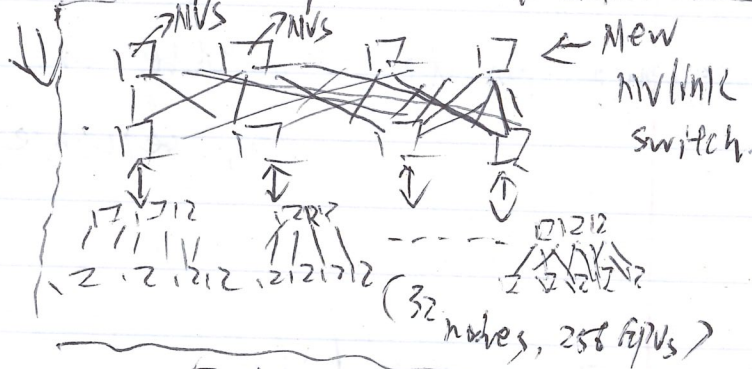
TPU networking toroidal mesh



GPU networking All-to-All (up to up 256)

PGX H100 256 superpod

only to connect with neighboring GPU



Fully connected - NVLink
massive bisection bandwidth.

part #1 recap:

New unit of compute - the data center.

what we want from multi-machine scaling

$\Rightarrow$ linear memory scaling (max model param scales with num GPUs)

$\Rightarrow$ linear compute scaling (model flops scale linearly with num GPUs)

Simple collective communication primitives